

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

1. Project Definition

Introduction

A **project** is a planned set of activities carried out to achieve a specific objective within a defined **scope, time, cost, and quality**.

In medical device development, a project may involve designing, developing, testing, validating, obtaining regulatory clearance/approval, and launching a new medical device.

Example

Development of an **IoT-based patient monitoring system** can be considered a project with:

- Objective – Develop a real-time patient monitoring system
- Duration – 12 months
- Budget – ₹20 lakh
- Team – Biomedical engineers, software engineers, clinicians and marketing personnel
- Deliverable – Tested and validated prototype/product

Major elements of project definition

1. **Project title**
2. **Problem statement**
3. **Objectives**
4. **Scope**
5. **Deliverables**
6. **Target customers**
7. **Timeline**
8. **Budget**
9. **Resources**
10. **Quality requirements**
11. **Regulatory requirements**
12. **Risk management**

Importance

A clear project definition helps the team understand **what has to be developed, why it is required, who will use it, how much it will cost, and when it should be completed**.

MEDICAL DEVICE MARKETING
UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

2. Financial Targets

Financial targets are the **specific financial goals established for a product or project**.

They help determine whether the product is commercially viable.

Important financial targets

1. Development budget

Maximum amount allocated for product development.

2. Manufacturing cost

Cost required to manufacture one unit.

3. Selling price

Price at which the product is sold.

4. Revenue target

Expected income from product sales.

5. Profit target

Expected profit from the product.

6. Break-even point

The sales level at which total revenue equals total cost.

7. Return on Investment (ROI)

Measures the return generated from an investment.

$$\text{ROI} = (\text{Net Return} / \text{Investment}) \times 100$$

Example

Suppose:

- Manufacturing cost = ₹8,000/device
- Selling price = ₹15,000/device
- Expected sales = 1,000 devices

The company can estimate its revenue, contribution margin, operating costs, and expected profit before deciding whether to proceed with the project.

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

3. Physician Input

Introduction

Physician input refers to obtaining information, suggestions, feedback, and clinical requirements from doctors during medical product development.

Physicians are important stakeholders because they understand the **clinical problem, patient requirements, workflow, treatment procedures, and limitations of existing technologies.**

Importance of Physician Input

1. Identification of clinical problems

Physicians can identify problems that require technological solutions.

Example:

A physician may identify that existing patient monitoring systems provide information too late for early intervention.

2. Defining product requirements

Doctors can specify:

- Required parameters
- Accuracy
- Response time
- Clinical functionality
- Patient population
- Desired alerts

3. Product design

Physician feedback helps engineers develop a device suitable for actual clinical environments.

4. Usability

Doctors can evaluate:

- User interface
- Display
- Controls
- Workflow
- Ease of operation

5. Clinical validation

Physicians may participate in clinical evaluation and provide feedback regarding clinical performance.

6. Risk identification

Doctors can identify possible clinical hazards and inappropriate uses.

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

7. Product acceptance

A product developed with appropriate clinical input is more likely to be accepted by healthcare professionals.

Methods of Obtaining Physician Input

- Interviews
- Questionnaires
- Focus groups
- Clinical observations
- Product demonstrations
- Advisory panels
- Prototype evaluation
- Clinical studies

Example

For a **smart prosthetic limb**, engineers may develop the sensors and control system, while physicians and rehabilitation professionals can provide input regarding:

- Patient comfort
- Gait requirements
- Safety
- Socket fit
- Rehabilitation needs
- Clinical usability

4. FDA Approval

Introduction

The **U.S. Food and Drug Administration (FDA)** regulates medical devices marketed in the United States. FDA's Center for Devices and Radiological Health (CDRH) regulates firms that manufacture, repack, relabel, or import medical devices for the U.S. market.

An important point for students is that **not every medical device receives an "FDA approval."** Depending on the device and regulatory pathway, the product may be **510(k) cleared, De Novo classified, PMA approved, or exempt from a premarket submission.**

Objectives of FDA Regulation

FDA regulation aims to ensure that medical devices:

- Are reasonably safe
- Are effective for their intended use

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Meet applicable regulatory requirements
- Are appropriately manufactured
- Have appropriate labeling
- Continue to be monitored after marketing

Step 1 – Identify the Intended Use

The first step is to clearly define:

- What the device does
- Who will use it
- Which patients will use it
- What disease/condition it addresses
- Whether it diagnoses, treats, prevents, monitors, or supports a condition

The intended use and indications for use are important because they influence the device's classification and regulatory pathway.

Step 2 – Determine Device Classification

FDA generally classifies medical devices into:

Class I – Low risk

Generally subject to **General Controls**.

Examples may include certain low-risk devices.

Class II – Moderate risk

Subject to **General Controls + Special Controls**.

Class III – High risk

Subject to **General Controls + Premarket Approval (PMA)** in applicable cases.

FDA states that classification is risk-based, with Class I generally representing the lowest risk and Class III the highest.

Class	Risk	Main Controls
Class I	Low	General Controls
Class II	Moderate	General + Special Controls
Class III	High	General Controls + PMA in applicable cases

MEDICAL DEVICE MARKETING

Unit-01 Introduction

Introduction to Medical Device Marketing

Medical device marketing is the systematic process of identifying healthcare needs, developing suitable medical products, communicating their benefits, creating awareness among customers, and making the products available to hospitals, healthcare professionals, patients, and other users. Unlike ordinary consumer products, medical devices are associated with **patient safety, clinical effectiveness, regulatory requirements, reimbursement, and ethical responsibilities.**

The success of a medical device depends not only on its technical performance but also on whether it solves a genuine healthcare problem, satisfies the requirements of doctors and patients, receives necessary regulatory approvals, is economically viable, and can be purchased or reimbursed within the healthcare system.

Major activities in medical device marketing

1. **Identification of healthcare need** – Understanding an existing clinical or patient problem.
2. **Customer identification** – Identifying patients, doctors, hospitals, technicians, distributors, and other stakeholders.
3. **Market research** – Studying market size, competitors, customer requirements, and trends.
4. **Product development** – Designing a device that satisfies clinical and technical requirements.
5. **Regulatory approval** – Obtaining necessary permissions and certifications.
6. **Pricing** – Determining an appropriate and competitive price.
7. **Promotion** – Communicating the benefits and applications of the product.
8. **Distribution** – Making the product available through appropriate channels.
9. **Sales and follow-up** – Supporting customers after purchase.
10. **Reimbursement** – Understanding how healthcare providers and patients will pay for the product.

Course Progress (use before starting new lesson)

Lesson 1. Object Oriented programming Concepts

Lesson 2. Fundamental programming Constructs

Lesson 3. Access Specifiers

Lesson 4. Constructors

Lesson 5. Inheritance



Introduction

- Object-Oriented Programming is a programming paradigm that uses “Objects” and their interactions to design applications.
- It simplifies the software development and maintenance by providing some concepts.



Object Oriented Programming Concepts

- . Object
- . Class
- . Abstraction
- . Encapsulation
- . Inheritance
- . Polymorphism



Object

- . An object is defined as the Realworld Entity.
- . It consists of properties, tasks performed by objects.
- . It also defined as instance of a Class

Ex: Human

Properties ☐ Name, Colour, Height

Tasks Performed ☐ Walk(), Run(), Work(), Read()

Creating an object

Creating an object of the class is also called as instantiating an object. Here, we create an object called “s” and print the value of regno.

```
public class student1
{
    int regno=5;
    public static void main(String args[])
    {
        student1 s=new student1();
        System.out.println(s.regno);
    }
}
```